

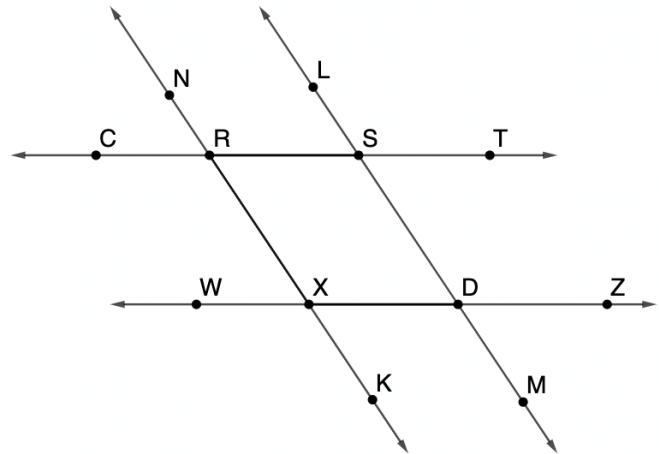
Geometry
Unit 3
Lesson 9 Practice

Name _____
Date _____
Period _____

Complete the two-column proof.

Given: $\overleftrightarrow{NK} \parallel \overleftrightarrow{LM}$ and $\angle LST \cong \angle WXK$

Prove: $\angle CRX \cong \angle RXD$



Statement	Reason
1. $\overleftrightarrow{NK} \parallel \overleftrightarrow{LM}$	1. Given
2.	2.
3.	3.
4.	4.
5.	5.
6.	6.

Write the statements and reasons in the correct order in the two-column proof.

$\angle CRX \cong \angle RXD$ Given

$\angle WXK \cong \angle XDM$ Transitive Property of Congruence

$\angle LST \cong \angle XDM$ If two parallel lines are intersected by a transversal, then the alternate interior angles are congruent.

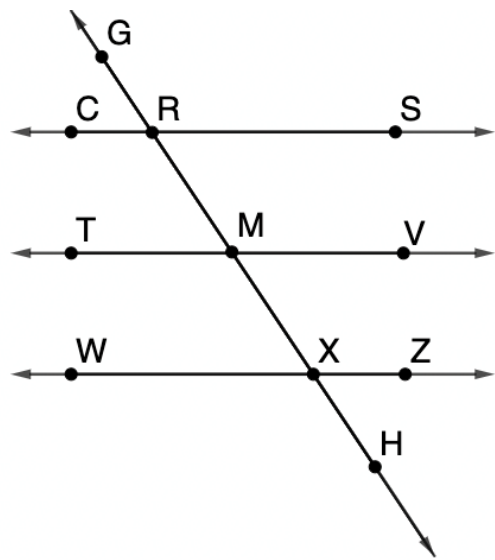
$\angle LST \cong \angle WXK$ If two parallel lines are intersected by a transversal, then the corresponding angles are congruent.

$\overleftrightarrow{CT} \parallel \overleftrightarrow{WZ}$ If alternate exterior angles are congruent, then the two lines that are intersected by a transversal are parallel.

Complete the two-column proof.

Given: $\angle RMV \cong \angle MXZ$ and
 $\angle GRC$ and $\angle WXH$ are supplementary

Prove: $\angle TMX \cong \angle GRS$



Statement	Reason
1. $\angle RMV \cong \angle MXZ$	1. Given
2.	2.
3. $\angle GRC$ and $\angle WXH$ are supplementary	3. Given
4.	4.
5.	5.
6.	6.
7.	7.

Write the statements and reasons in the correct order in the two-column proof.

$\overleftrightarrow{TV} \parallel \overleftrightarrow{WZ}$	Supplements of congruent angles are congruent.
$\overleftrightarrow{CS} \parallel \overleftrightarrow{WZ}$	If two angles form a linear pair, then they are supplementary.
$\angle TMX \cong \angle GRS$	If two parallel lines are intersected by a transversal, then the alternate exterior angles are congruent.
$\angle GRC \cong \angle MXW$	If two parallel lines are intersected by a transversal, then the corresponding angles are congruent.
$\angle MXW$ and $\angle WXH$ are supplementary	If corresponding angles are congruent, then the two lines that are intersected by a transversal are parallel.